Assignment
(Polymer)

19.1b Calculate the number-average molar mass and the mass-average molar mass of a mixture of two polymers, one having $M = 62\ \mathrm{kg\ mol^{-1}}$ and the other $M = 78\ \mathrm{kg\ mol^{-1}}$, with their amounts (numbers of moles) in the ratio 3:2.

19.2b The radius of gyration of a long chain molecule is found to be 18.9 nm. The chain consists of links of length 450 pm. Assume the chain is randomly coiled and estimate the number of links in the chain.

19.3b A solution consists of 25 per cent by mass of a trimer with $M = 22\ \mathrm{kg\ mol^{-1}}$ and its monomer. What average molar mass would be obtained from measurement of: (a) osmotic pressure, (b) light scattering?

19.11b A polymer chain consists of 1200 segments, each 1.125 nm long. If the chain were ideally flexible, what would be the r.m.s. separation of the ends of the chain?

19.12b Calculate the contour length (the length of the extended chain) and the root mean square separation (the end-to-end distance) for polypropylene of molar mass $174\ \mathrm{kg\ mol^{-1}}$.